

5

Motion in step potentials

We follow up our study of the harmonic oscillator by looking at motion in a wider range of one-dimensional potentials $V(x)$. The potentials we study will be artificial in that they will only vary in sharp steps, but they will enable us to explore analytically some features of quantum mechanics that are generic and hidden from us in the classical limit. We start by considering a particle that is trapped in a potential well and go on to consider a particle that has a choice of two wells. We find that in this case it can move between these wells in violation of classical mechanics, and we use this simple system to model the operation of an ammonia maser. In §5.3 we ask how potential wells and barriers affect the motion of a free particle – one that can escape to infinity. We find that whereas in classical mechanics the particle is never reflected by a potential well, in quantum mechanics there is generally a non-zero amplitude for such reflection. We find also that particles can ‘tunnel’ through barriers that classically would certainly reflect them.

5.1 Square potential well

We look for energy eigenstates of a particle that moves in the potential (Figure 5.1)

$$V(x) = \begin{cases} 0 & \text{for } |x| < a \\ V_0 > 0 & \text{otherwise.} \end{cases} \quad (5.1)$$

Since V is an even function of x , the Hamiltonian (2.51) commutes with the parity operator P (page 77). So there is a complete set of energy

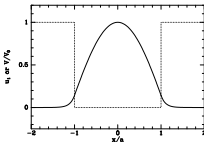


Figure 5.1 The dotted line shows the square-well potential $V(x)$. The full curve shows the ground-state wavefunction.

eigenstates of well-defined parity. The wavefunctions $u(x) = \langle x|E \rangle$ of these states will be either even or odd functions of x , and this fact will greatly simplify the job of determining $u(x)$.

In the position representation, the governing equation (the TISE 2.33) reads

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dx^2} + V(x)u = Eu. \quad (5.2)$$

On account of the step-like nature of V , equation (5.2) reduces to a pair of extremely simple equations,

$$\begin{aligned} \frac{d^2 u}{dx^2} &= -\frac{2mE}{\hbar^2} u & \text{for } |x| < a \\ \frac{d^2 u}{dx^2} &= \frac{2m(V_0 - E)}{\hbar^2} u & \text{otherwise.} \end{aligned} \quad (5.3)$$

We restrict ourselves to solutions that describe a particle that is bound by the potential well in the sense that $E < V_0$.¹ Then the solution to the second equation is $u(x) = Ae^{\pm Kx}$, where A is a constant and

$$K = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}. \quad (5.4)$$

If u is to be normalisable, it must vanish as $|x| \rightarrow \infty$. So at $x > a$ we have $u(x) = Ae^{-Kx}$, and at $x < -a$ we have $u(x) = \pm Ae^{+Kx}$, where the plus sign is required for solutions of even parity, and the minus sign is required for odd parity.

For $E > 0$, the solution to the first of equations (5.3) is either $u(x) = B \cos(kx)$ or $u(x) = B \sin(kx)$ depending on the parity, where

$$k = \sqrt{\frac{2mE}{\hbar^2}}. \quad (5.5)$$

¹ By considering the behaviour of u near the origin we can prove that $E > 0$.

So far we have ensured that $u(x)$ solves the TISE everywhere except at $|x| = a$. Unless u is continuous at these points, du/dx will be arbitrarily large, and d^2u/dx^2 will be undefined, so u will not satisfy the TISE. Similarly, unless du/dx is continuous at these points, d^2u/dx^2 will be arbitrarily large, so u cannot solve the TISE. Therefore, we require that both u and du/dx are continuous at $x = a$, that is

$$\left. \begin{aligned} B \cos(ka) &= Ae^{-Ka} \\ -kB \sin(ka) &= -KAe^{-Ka} \end{aligned} \right\} \quad \text{or} \quad \left\{ \begin{aligned} B \sin(ka) &= Ae^{-Ka} \\ kB \cos(ka) &= -KAe^{-Ka} \end{aligned} \right. \quad (5.6)$$

where the first pair of equations apply in the case of even parity and the second in the case of odd parity. It is easy to show that once these equations have been satisfied, the corresponding equations for $x = -a$ will be automatically satisfied.

We eliminate A and B from equations (5.6) by dividing the second equation in each set by the first. In the case of even parity we obtain

$$k \tan(ka) = K = \sqrt{\frac{2mV_0}{\hbar^2} - k^2}. \quad (5.7)$$

This is an algebraic equation for k , which controls E through (5.5). Before attempting to solve this equation, it is useful to rewrite it as

$$\tan(ka) = \sqrt{\frac{W^2}{(ka)^2} - 1} \quad \text{where} \quad W = \sqrt{\frac{2mV_0a^2}{\hbar^2}}. \quad (5.8)$$

W and ka are dimensionless variables. The left and right sides of equation (5.8) are plotted as functions of ka in Figure 5.2. Since for $ka = 0$ the graphs of the two sides start at the origin and infinity, and the graph of the left side increases to infinity at $ka = \pi/2$ while the graph of the right side terminates at $ka = W$, the equation always has at least one solution. Thus no matter how small V_0 and a are, the square well can always trap the particle. The bigger W is, the more solutions the equation has; a second solution appears at $W = \pi$, a third at $W = 2\pi$, etc.

Analogously one can show that for an odd-parity energy eigenstate to exist, we must have $W > \pi/2$ and that additional solutions appear when $W = (2r+1)\pi/2$ for $r = 1, 2, \dots$ (Problem 5.5).

From a naive perspective our discovery that no matter how narrow or shallow it is, a square potential well always has at least one bound state, conflicts with the uncertainty principle: the particle's momentum cannot exceed $p_{\max} = \sqrt{2mE} < \sqrt{2mV_0}$ and can have either sign, so if the particle were confined within the well, the product of the uncertainties in p and x would be less than $4ap_{\max} < 4\sqrt{2mV_0}a^2 = 4\hbar W$, which tends to zero with W . The resolution of this apparent paradox is that for $W \ll 1$ the particle is not confined *within* the well; there is a non-negligible probability of finding the particle in the classically forbidden region $|x| > a$. In the limit $W \rightarrow 0$ the particle is confined by a well in which it is certain never to be found!

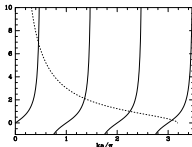


Figure 5.2 Plots of the left (full) and right (dashed) sides of equation (5.8) for the case $W = 10$.



Figure 5.3 A square well inscribed in a general well.

Our result that a square well always has a bound state can be extended to potential wells of any shape: given the potential well U sketched in Figure 5.3, we consider the square well shown by the dashed line in the figure. Since this shallower and narrower well has a bound state, we infer that the potential U also has at least one bound state.

5.1.1 Limiting cases

Infinitely deep well It is worthwhile to investigate the behaviour of these solutions as $V_0 \rightarrow \infty$ with a fixed, when the well becomes infinitely deep. Then $W \rightarrow \infty$ and the dashed curve in Figure 5.2 moves higher and higher up the paper and hits the x -axis further and further to the right. Consequently, the values of ka that solve equation (5.8) tend towards $ka = (2r + 1)\pi/2$, so the even-parity eigenfunctions become

$$u(x) = \begin{cases} A \cos[(2r + 1)\pi x/2a] & |x| < a \\ 0 & \text{otherwise.} \end{cases} \quad (5.9)$$

This solution has a discontinuity in its gradient at $x = a$ because it is the limit of solutions in which the curvature K for $x > a$ diverges to infinity. The odd-parity solutions are obtained by replacing the cosine with $\sin(s\pi x/a)$, where $s = 1, 2, \dots$, which again vanish at the edge of the well

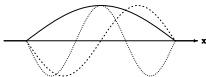


Figure 5.4 The wavefunctions of the lowest three stationary states of the infinitely deep square well: ground state (full); first excited state (dashed); second excited state (dotted).

(Figure 5.4). From this example we infer the principle that *wavefunctions vanish at the edges of regions of infinite potential energy*.

The energy of any stationary state of an infinite square potential well can be obtained from

$$E_n = \frac{n^2}{8m} \left(\frac{\hbar\pi}{a} \right)^2, \quad \text{where } n = 1, 2, \dots \quad (5.10)$$

The particle's momentum when it is in the ground state ($n = 1$) is of order $\hbar k = \hbar\pi/2a$ and of undetermined sign, so the uncertainty in the momentum is $\Delta_p \simeq \hbar\pi/a$. The uncertainty in the particle's position is $\Delta_x \simeq 2a$, so $\Delta_x \Delta_p \simeq 2\hbar\pi$, consistent with the uncertainty principle (§2.3.2).

Infinitely narrow well In §12.5.1 we will study a model of covalent bonding that involves the potential obtained by letting the width of the square well tend to zero as the well becomes deeper and deeper in such a way that the product $V_0 a$ remains constant. In this limit $W \propto a\sqrt{V_0}$ (eq. 5.8) tends to zero, so there is only one bound state and it will be an even-parity state.

Rather than obtaining the wavefunction and energy of this state from formulae already in hand, it is more convenient to reformulate the problem using a different normalisation for the energy: we now set V to zero outside the well, so V becomes negative at interior points. Then we can write $V(x) = -V_\delta \delta(x)$, where $\delta(x)$ is the Dirac delta function and $V_\delta > 0$. The TISE now reads

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dx^2} - V_\delta \delta(x) u = E u. \quad (5.11)$$

Integrating the equation from $x = -\epsilon$ to $x = \epsilon$ with ϵ infinitesimal, we find

$$-\left[\frac{du}{dx} \right]_{-\epsilon}^{\epsilon} = \frac{2m}{\hbar^2} \left(V_\delta u(0) + E \int_{-\epsilon}^{\epsilon} dx u \right). \quad (5.12)$$

Since u is finite, the integral on the right can be made as small as we please by letting $\epsilon \rightarrow 0$. Hence the content of equation (5.12) is that du/dx has a discontinuity at the origin:

$$\left[\frac{du}{dx} \right]_{-\epsilon}^{\epsilon} = -\frac{2mV_\delta}{\hbar^2} u(0). \quad (5.13)$$

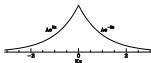


Figure 5.5 Wavefunction of a particle trapped by a very narrow, deep potential well.

Since we know that the solution we seek has even parity, it is of the form $u(x) = Ae^{\mp Kx}$, where the minus sign applies for $x > 0$ (Figure 5.5). Substituting this form of u into (5.13) and dividing through by $2A$ we have

$$K = \frac{mV_\delta}{\hbar^2}. \quad (5.14)$$

Inserting $u = e^{-Kx}$ into equation (5.11) at $x > 0$ we find that $E = -\hbar^2 K^2/2m$, so the energy of a particle that is bound to a δ -function potential is

$$E = -\frac{mV_\delta^2}{2\hbar^2}. \quad (5.15)$$

Figure 5.5 shows that $|\psi(x)|^2$ is finite in the well, and the well is infinitely narrow, so the probability of finding the particle in the well is zero – the particle is certain never to be in the well that traps it! This result is an extreme case of the phenomenon we discussed apropos the application of the uncertainty principle to a shallow well of finite depth.

5.2 A pair of square wells

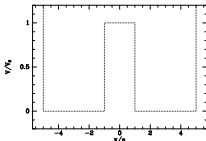
Some important phenomena can be illustrated by considering motion in a pair of potentials that are separated by a barrier of finite height and width. Figure 5.6 shows the potential

$$V(x) = \begin{cases} V_0 & \text{for } |x| < a \\ 0 & \text{for } a < |x| < b \\ \infty & \text{otherwise.} \end{cases} \quad (5.16)$$

Since the potential is an even function of x , we may assume that the energy eigenfunctions that we seek are of well-defined parity.

For simplicity we take the potential to be infinite for $|x| > b$, and we assume that the particle is classically forbidden in the region $|x| < a$. Then in this region the wavefunction must be of the form $u(x) = A \cosh(Kx)$ or $u(x) = A \sinh(Kx)$ depending on parity, and K is given by (5.4). In the region $a < x < b$ the wavefunction may be taken to be of the form $u(x) = B \sin(kx + \phi)$, where B and ϕ are constants to be determined and k is related to the energy by (5.5). From our study of a single square well we know that u must vanish at $x = b$, so

$$\sin(kb + \phi) = 0 \quad \Rightarrow \quad \phi = r\pi - kb \quad \text{with } r = 0, 1, \dots \quad (5.17)$$

Figure 5.6 A double potential well with $b/a = 5$.

Again by analogy with the case of a single square well, we require u and its derivative to be continuous at $x = a$, so (depending on parity)

$$\left. \begin{aligned} \cosh(Ka) &= B \sin(ka + \phi) \\ K \sinh(Ka) &= kB \cos(ka + \phi) \end{aligned} \right\} \text{ or } \left\{ \begin{aligned} \sinh(Ka) &= B \sin(ka + \phi) \\ K \cosh(Ka) &= kB \cos(ka + \phi) \end{aligned} \right. \quad (5.18)$$

Once these equations have been solved, the corresponding conditions at $x = -a$ will be automatically satisfied if for $-b < x < -a$ we take $u = \pm B \sin(k|x| + \phi)$, using the plus sign in the even-parity case.

Using (5.17) to eliminate ϕ from equations (5.18) and then proceeding in close analogy with the working below equations (5.6), we find

$$\tan[r\pi - k(b-a)] \sqrt{\frac{W^2}{(ka)^2} - 1} = \begin{cases} \coth\left(\sqrt{W^2 - (ka)^2}\right) & \text{even parity} \\ \tanh\left(\sqrt{W^2 - (ka)^2}\right) & \text{odd parity,} \end{cases} \quad (5.19)$$

where W is defined by equation (5.8).

The left and right sides of equation (5.19) are plotted in Figure 5.7; the values of ka for stationary states correspond to intersections of the steeply sloping curves of the left side with the initially horizontal curves of the right side. The smallest value of ka is associated with the ground state. The values come in pairs, one for an even-parity state, and one very slightly larger for an odd-parity state. The difference between the k values in a pair increases with k .

The closeness of the k values in a given pair ensures that in the right-hand well ($a < x < b$) the wavefunctions $u_e(x)$ and $u_o(x)$ of the even- and odd-parity states are very similar, and that in the left-hand well u_e and u_o differ by little more than sign – see Figure 5.8. Moreover, when the k values are similar, the amplitude of the wavefunction is small in the classically forbidden region $|x| < a$. Hence, the linear combinations

$$\psi_{\pm}(x) = \frac{1}{\sqrt{2}} [u_e(x) \pm u_o(x)] \quad (5.20)$$

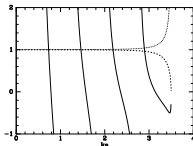


Figure 5.7 Full curve: the left side of equation (5.19) for the case $W = 3.5$, $b = 5a$. Each vertical section is associated with a different value of the integer r . The right side is shown by the dotted curve for even parity, and the dashed curve for odd parity.

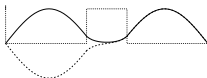


Figure 5.8 The ground state (full curve) and the associated odd-parity state (dashed curve) of the double square-well potential (shown dotted).

are the wavefunctions of a state $|\psi_+\rangle$ in which the particle is almost certain to be in the right-hand well, and a state $|\psi_-\rangle$ in which it is equally certain to be in the left-hand well.

Consider now how the system evolves if at time 0 it is in the state $|\psi_+\rangle$, so the particle is in the right-hand well. Then by equation (2.32) at time t its wavefunction is

$$\begin{aligned}\psi(x, t) &= \frac{1}{\sqrt{2}} \left[u_e(x) e^{-iE_e t/\hbar} + u_o(x) e^{-iE_o t/\hbar} \right] \\ &= \frac{1}{\sqrt{2}} e^{-iE_e t/\hbar} \left[u_e(x) + u_o(x) e^{-i(E_o - E_e)t/\hbar} \right].\end{aligned}\quad (5.21)$$

After a time $T = \pi\hbar/(E_o - E_e)$ the exponential in the square brackets on the second line of this equation equals -1 , so to within an overall phase factor the wavefunction has become $[u_e(x) - u_o(x)]/\sqrt{2}$, implying that the particle is certainly in the left-hand well; we say that in the interval T the particle has **tunnelled** through the barrier that divides the wells. After a further period T it is certainly in the right-hand well, and so on ad

infinitum. In classical physics the particle would stay in whatever well it was in initially. In fact, the position of a familiar light switch is governed by a potential that consists of two similar adjacent potential wells, and such switches most definitely do not oscillate between their on and off positions. We do not observe tunnelling in the classical regime because $E_0 - E_w$ decreases with increasing W faster than e^{-2W} (Problem 5.16), so the time required for tunnelling to occur increases faster than e^{2W} and is enormously long for classical systems such as light switches.

5.2.1 Ammonia

Nature provides us with a beautiful physical realisation of a system with a double potential well in the ammonia molecule NH_3 . Ammonia contains four nuclei and ten electrons, so is really a very complicated dynamical system. However, in §12.5.2 we shall show that a useful way of thinking about the low-energy behaviour of molecules is to imagine that the electrons provide light springs, which hold the nuclei together. The nuclei oscillate around the equilibrium positions defined by the potential energy of these springs. In the case of NH_3 , the potential energy is minimised when the three hydrogen atoms are arranged at the vertices of an equilateral triangle, while the nitrogen atom lies some distance x away from the plane of the triangle, either 'above' or 'below' it (see Figure 5.9). Hence if we were to plot the molecule's potential energy as a function of x , we would obtain a graph that looked like Figure 5.6 except that the sides of the wells would be sloping rather than straight. This function would yield eigenenergies that came in pairs, as in our square-well example.

In many physical situations the molecule would have so little energy that it could have negligible amplitudes to be found in any but the two lowest-lying stationary states, and we would obtain an excellent approximation to the dynamics of ammonia by including only the amplitudes to be found in these two states. We now use Dirac notation to study this dynamics.

Let $|+\rangle$ be the state whose wavefunction is analogous to the wavefunction $\psi_+(x)$ defined above in the case of the double square well; then $\psi_+(x) = \langle x|+\rangle$, and in the state $|+\rangle$ the N atom is certainly above the plane containing the H atoms. The ket $|-\rangle$ is the complementary state in which the N atom lies below the plane of the H atoms.

The $|\pm\rangle$ states are linear combinations of the eigenkets $|e\rangle$ and $|o\rangle$ of the Hamiltonian:

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|e\rangle \pm |o\rangle). \quad (5.22)$$

In the $|\pm\rangle$ basis the matrix elements of the Hamiltonian H are

$$\begin{aligned} \langle +|H|+ \rangle &= \frac{1}{2}(\langle e| + \langle o|)H(|e\rangle + |o\rangle) = \frac{1}{2}(E_e + E_o) \\ \langle +|H| - \rangle &= \frac{1}{2}(\langle e| + \langle o|)H(|e\rangle - |o\rangle) = \frac{1}{2}(E_e - E_o) \\ \langle -|H| - \rangle &= \frac{1}{2}(\langle e| - \langle o|)H(|e\rangle - |o\rangle) = \frac{1}{2}(E_e + E_o). \end{aligned} \quad (5.23)$$

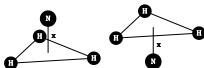


Figure 5.9 The two possible relative locations of nitrogen and hydrogen atoms in NH_3 .

Bearing in mind that H is represented by a Hermitian matrix, we conclude that it is

$$H = \begin{pmatrix} \bar{E} & -A \\ -A & \bar{E} \end{pmatrix}, \quad (5.24)$$

where $\bar{E} = \frac{1}{2}(E_0 + E_1)$ and $A = \frac{1}{2}(E_0 - E_1)$ are both positive.

Now the electronic structure of NH_3 is such that the N atom carries a small negative charge $-q$, with a corresponding positive charge $+q$ distributed among the H atoms. With NH_3 in either the $|+\rangle$ or $|-\rangle$ state there is a net separation of charge, so an ammonia molecule in these states possesses an electric dipole moment of magnitude qs directed perpendicular to the plane of H atoms (see Figure 5.9), where s is a small distance.

Below equation (5.21) we saw that a molecule that is initially in the state $|+\rangle$ will subsequently oscillate between this state and the state $|-\rangle$ at a frequency $(E_0 - E_1)/2\pi\hbar = A/\pi\hbar$. Hence a molecule that starts in the state $|+\rangle$ is an oscillating dipole and it will emit electromagnetic radiation at the frequency $A/\pi\hbar$. This proves to be 150 GHz, so the molecule emits microwave radiation.

The ammonia maser The energy $2A$ that separates the ground and first excited states of ammonia in zero electric field is small, 10^{-4} eV. Consequently at room temperature similar numbers of molecules are in these two states. The principle of an ammonia maser² is to isolate the molecules that are in the first excited state, and then to harvest the radiation that is emitted as the molecules decay to the ground state. The isolation is achieved by exploiting the fact that, as we now show, when an electric field is applied, molecules in the ground and first excited states develop polarisations of opposite sign.

We define the dipole-moment operator P by

$$P|+\rangle = -qs|+\rangle, \quad P|-\rangle = +qs|-\rangle, \quad (5.25)$$

so a molecule in the $|+\rangle$ state has dipole moment $-qs$ and a molecule in the $|-\rangle$ state has dipole moment $+qs$.³ To measure this dipole moment, we can place the molecule in an electric field of magnitude $\mathcal{E}$ parallel to

² 'maser' is an acronym for 'microwave amplification by stimulated emission of radiation'.

³ The N atom is negatively charged so the dipole points away from it.

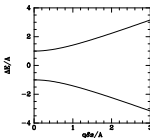


Figure 5.10 Energy levels of the ammonia molecule as a function of external electric field strength $\mathcal{E}$. The quantity plotted is $\Delta E = E - \bar{E}$.

the dipole axis. Since the energy of interaction between a dipole P and an electric field $\mathcal{E}$ is $-P\mathcal{E}$, the new Hamiltonian is

$$H = \begin{pmatrix} \bar{E} + q\mathcal{E}s & -A \\ -A & \bar{E} - q\mathcal{E}s \end{pmatrix}. \quad (5.26)$$

This new Hamiltonian has eigenvalues

$$E_{\pm} = \bar{E} \pm \sqrt{A^2 + (q\mathcal{E}s)^2}. \quad (5.27)$$

These are plotted as a function of field $\mathcal{E}$ in Figure 5.10. When $\mathcal{E} = 0$ the energy levels are the same as before. As $\mathcal{E}$ slowly increases, E increases quadratically with $\mathcal{E}$, because $\sqrt{A^2 + (q\mathcal{E}s)^2} \simeq A + (q\mathcal{E}s)^2/2A$, but when $\mathcal{E} \gg A/qs$ the energy eigenvalues change linearly with $\mathcal{E}$. Notice that in this large-field limit, at lowest order the energy levels do not depend on A .

The physical interpretation of these results is the following. In the absence of an electric field, the energy eigenstates are the states of well-defined parity $|e\rangle$ and $|o\rangle$, which have no dipole moment. An electric field breaks the symmetry between the two potential wells, making it energetically favourable for the N atom to occupy the well to which the electric field is pushing it. Consequently, the ground state develops a dipole moment P , which is proportional to $\mathcal{E}$. Thus at this stage the electric contribution to the energy of the ground state, which is $-P\mathcal{E}$, is proportional to $\mathcal{E}^2$. Once this contribution exceeds the separation A between the states of well-defined parity, the molecule has shifted to the lower-energy state of the pair $|\pm\rangle$, and it stays in this state as the electric field is increased further. Thus for large fields the polarisation of the ground state is independent of $\mathcal{E}$ and the electric contribution to the energy is simply proportional to $\mathcal{E}$.

While the ground state develops a dipole moment that lowers its energy, the first excited state develops the opposite polarisation, so the

electric field raises the energy of this state, as shown in Figure 5.10. The response of the first excited state is anomalous from a classical perspective.

Ehrenfest's theorem (2.57) tells us that the expectation values of operators obey classical equations of motion. In particular the momentum of a molecule obeys

$$\frac{d\langle p_x \rangle}{dt} = - \left\langle \frac{\partial V}{\partial x} \right\rangle, \quad (5.28)$$

where x is a Cartesian coordinate of the molecule's centre of mass. The potential depends on x only through the electric field $\mathcal{E}$, so

$$\frac{\partial V}{\partial x} = -P \frac{\partial \mathcal{E}}{\partial x}, \quad (5.29)$$

from which it follows that

$$\frac{d\langle p_x \rangle}{dt} = \langle P \rangle \frac{\partial \mathcal{E}}{\partial x}. \quad (5.30)$$

Since the sign of $\langle P \rangle$ and therefore the force on a molecule depends on whether the molecule is in the ground or first excited state, when a jet of ammonia passes through a region of varying $\mathcal{E}$, molecules in the first excited state can be separated from those in the ground state.

Having gathered the molecules that are in the excited state, we lead them to a cavity that resonates with the 150 GHz radiation that is emitted when molecules drop into the ground state. The operation of an ammonia maser by Charles Townes and colleagues⁴ was the first demonstration of stimulated emission and opened up the revolution in science and technology that lasers have since wrought.

5.3 Scattering of free particles

We now consider what happens when a particle that is travelling parallel to the x -axis encounters a region of sharply changed potential energy. In classical physics the outcome depends critically on whether the potential rises by more than the kinetic energy of the incoming particle: if it does, the particle is certainly reflected, while it continues moving towards positive x in the contrary case. We shall find that quantum mechanics predicts that there are usually non-vanishing probabilities for both reflection and transmission regardless of whether the rise in potential exceeds the initial kinetic energy.

We assume that each particle has well-defined energy E , so its wavefunction satisfies the TISE (5.2). We take the potential to be (Figure 5.11)

$$V = \begin{cases} V_0 & \text{for } |x| < a \\ 0 & \text{otherwise,} \end{cases} \quad (5.31)$$

⁴ J.P. Gordon, H.J. Zeiger, and C.H. Townes, *Phys. Rev.* **95**, 282 (1954).

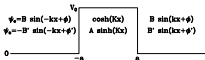


Figure 5.11 A square, classically forbidden barrier and the functional forms for stationary states of even (top) and odd parity.

where V_0 is a constant. At $|x| > a$ the relevant solutions of (5.2) are

$$e^{\pm ikx} \quad \text{or} \quad \begin{cases} \sin(kx + \phi) & \text{at } x > a \\ \pm \sin(-kx + \phi) & \text{at } x < -a \end{cases} \quad \text{with } k = \sqrt{\frac{2mE}{\hbar^2}}, \quad (5.32a)$$

where ϕ is a constant phase. Since the time dependence of these stationary states is obtained by introducing the factor $e^{-iEt/\hbar}$, a plus sign in $e^{\pm ikx}$ implies that the particle is moving to the right, and a minus sign is associated with movement to the left. A wavefunction that is proportional to $\sin(kx + \phi)$ contains both types of wave with amplitudes of equal magnitude, so it makes motion in either direction equally likely. At $|x| < a$ the relevant solutions of (5.2) are

$$e^{\pm iKx} \quad \text{or} \quad \begin{cases} \cos(Kx) \\ \sin(Kx) \end{cases} \quad \text{with } K = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \quad \text{when } E > V_0$$

$$e^{\pm Kx} \quad \text{or} \quad \begin{cases} \cosh(Kx) \\ \sinh(Kx) \end{cases} \quad \text{with } K = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \quad \text{when } E < V_0. \quad (5.32b)$$

In every case we have a choice between exponential solutions and solutions of well-defined parity. Since our physical problem is strongly asymmetric in that particles are fired in from negative x rather than equally from both sides, it is tempting to work with the exponential solutions of the TISE rather than the solutions of well-defined parity. However, the algebra involved in solving our problem is much lighter if we use solutions of well-defined parity because then the conditions that ensure proper behaviour of the solution at $x = a$ automatically ensure that the solution also behaves properly at $x = -a$; if we use exponential solutions, we have to deal with the cases $x = \pm a$ individually. Therefore we seek solutions of the form

$$\psi_e(x) = \begin{cases} B \sin(k|x| + \phi) & \text{for } |x| > a \\ \cos(Kx) \text{ or } \cosh(Kx) & \text{otherwise;} \end{cases}$$

$$\psi_o(x) = \begin{cases} B' \sin(kx + \phi') & \text{for } x > a \\ A \sin(Kx) \text{ or } A \sinh(Kx) & \text{for } |x| \leq a, \\ -B' \sin(k|x| + \phi') & \text{otherwise,} \end{cases} \quad (5.33)$$

where A, B, B', ϕ and ϕ' are constants. B, ϕ and ϕ' will be unambiguously determined by the conditions at $x = \pm a$. These conditions will make B' proportional to A , which we treat as a free parameter.

In our study of the bound states of potential wells in §5.1, the requirement that the wavefunction vanish at infinity could be satisfied only for discrete values of E . These values of E (and therefore k and K) differed between the even- and odd-parity solutions, so all energy eigenfunctions automatically had well-defined parity. In the case of a free particle, by contrast, we will be able to construct both an even-parity and an odd-parity solution for every given value of E . Linear combinations of these solutions $\psi_e(x)$ and $\psi_o(x)$ of well-defined parity are energy eigenfunctions that do not have well-defined parity. We now show that the sum these solutions of the TISE with well-defined parity can be made to describe the actual scattering problem.

In the solution we seek, there are no particles approaching from the right. Adding the even- and odd-parity solutions, we obtain at $x > a$ a solution of the form

$$\begin{aligned}\psi_e(x) + \psi_o(x) &= B \sin(kx + \phi) + B' \sin(kx + \phi') \\ &= \frac{e^{ikx}}{2i} (Be^{i\phi} + B'e^{i\phi'}) - \frac{e^{-ikx}}{2i} (Be^{-i\phi} + B'e^{-i\phi'}).\end{aligned}\quad (5.34)$$

The condition that no particles are approaching from the right is

$$B' = -Be^{i(\phi' - \phi)}, \quad (5.35)$$

for then at $x > a$ the solution becomes

$$\psi_e(x) + \psi_o(x) = \frac{e^{ikx}}{2i} Be^{i\phi} (1 - e^{2i(\phi' - \phi)}) \quad (x > a), \quad (5.36)$$

which includes only particles moving to the right. At $x < -a$ the solution is now

$$\begin{aligned}\psi_e(x) + \psi_o(x) &= B \sin(-kx + \phi) - B' \sin(-kx + \phi') \\ &= \frac{e^{ikx}}{2i} (-Be^{-i\phi} + B'e^{-i\phi'}) + \frac{e^{-ikx}}{2i} (Be^{i\phi} - B'e^{i\phi'}) \\ &= e^{ikx} i Be^{-i\phi} + \frac{e^{-ikx}}{2i} Be^{i\phi} (1 + e^{2i(\phi' - \phi)}).\end{aligned}\quad (5.37a)$$

In the solution given by equations (5.36) and (5.37a) the incoming amplitude is $iBe^{-i\phi}$, while the amplitudes for reflection and transmission are

$$\begin{aligned}\frac{B}{2i} e^{i\phi} (1 + e^{2i\Delta\phi}) &\quad (\text{reflected}) \\ \frac{B}{2i} e^{i\phi} (1 - e^{2i\Delta\phi}) &\quad (\text{transmitted}),\end{aligned}\quad (5.37b)$$

where

$$\Delta\phi = \phi' - \phi \quad (5.37c)$$

is the **phase difference** $\Delta\phi$ between the odd- and even-parity solutions at $|x| > a$. From the ratios of the mod-squares of the outgoing amplitudes

to that of the incoming amplitude iB we have that the reflection and transmission probabilities are

$$P_{\text{ref}} = \cos^2(\Delta\phi) \quad P_{\text{trans}} = \sin^2(\Delta\phi). \quad (5.38)$$

Thus $\Delta\phi$ determines the reflection and transmission probabilities. Notice that these formulae for the transmission and reflection probabilities have been obtained without reference to the form of the wavefunction at $|x| < a$. Consequently, they are valid for any scattering potential $V(x)$ that has even parity and vanishes outside some finite region, here $|x| < a$.

The scattering cross-section In the case that $V_0 < 0$, so the scattering potential forms a potential well, the outgoing wave at $x > a$ represents two physically distinct possibilities: (i) that the incoming particle failed to interact with the potential well and continued on its way undisturbed, and (ii) that it was for a while trapped by the well and later broke free towards the right rather than the left. We isolate the possibility of scattering by writing the amplitude of the outgoing wave as $1 + T$ times the amplitude of the incoming wave. Here the one represents the possibility of passing through undisturbed and T represents real forward scattering. From our formulae (5.37) for the amplitudes of the incoming and outgoing waves we have that

$$T = \frac{1}{2} \left(e^{2i\phi'} - e^{2i\phi} \right) - 1. \quad (5.39a)$$

If we similarly write the amplitude of the reflected wave as R times the amplitude of the incoming wave, then from the formulae above we have

$$R = -\frac{1}{2} \left(e^{2i\phi'} + e^{2i\phi} \right). \quad (5.39b)$$

The **total scattering cross-section**⁵ is defined to be the sum of the probabilities for forward and backward scattering:

$$\sigma = |R|^2 + |T|^2. \quad (5.40)$$

Now $|R|^2$ is just the reflection probability P_{ref} , and the transmission probability is

$$P_{\text{trans}} = |1 + T|^2 = 1 + |T|^2 + T + T^*, \quad (5.41)$$

so

$$\sigma = P_{\text{ref}} + P_{\text{trans}} - 1 - T - T^* = -(T + T^*). \quad (5.42)$$

From equation (5.39a) we have an expression for the total scattering cross-section in terms of the phase angles

$$\sigma = 2 - \cos(2\phi') + \cos(2\phi). \quad (5.43a)$$

The trigonometric identities $1 + \cos 2\phi = 2 \cos^2 \phi$ and $1 - \cos 2\phi = 2 \sin^2 \phi$ enable us to re-express the cross-section as

$$\sigma = 2 \left(\sin^2 \phi' + \cos^2 \phi \right). \quad (5.43b)$$

⁵ This definition of the total scattering cross-section only applies to one-dimensional scattering problems. See §13.3 for the definition of the total scattering cross-section that is appropriate for realistic three-dimensional experiments.

5.3.1 Tunnelling through a potential barrier

Now consider the case $V_0 > E$ in which classical physics predicts that all particles are reflected. From equations (5.33), the conditions for both the wavefunction and its derivative to be continuous at $x = a$ are

$$\left. \begin{aligned} \cosh(Ka) &= B \sin(ka + \phi) \\ K \sinh(Ka) &= Bk \cos(ka + \phi) \end{aligned} \right\} \text{ or } \left\{ \begin{aligned} A \sinh(Ka) &= B' \sin(ka + \phi') \\ KA \cosh(Ka) &= B'k \cos(ka + \phi') \end{aligned} \right. \quad (5.44a)$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}} \quad \text{and} \quad K = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}. \quad (5.44b)$$

Dividing the equations of each pair into one another to eliminate the constants A, B and B' , we obtain

$$\tan(ka + \phi) = (k/K) \coth(Ka) \quad \text{or} \quad \tan(ka + \phi') = (k/K) \tanh(Ka). \quad (5.45)$$

On account of the fact that for any x , $\tan(x + \pi) = \tan x$, the equations have infinitely many solutions for ϕ and ϕ' that differ by $r\pi$, where r is an integer. From equations (5.39) and (5.43a) we see that these solutions give identical amplitudes for reflection and transmission and the same value of the total scattering cross-section σ . Hence we need consider only the unique values of ϕ and ϕ' that lie within $\pm\pi/2$ of $-ka$.

Equations (5.38) show that the transmission and reflection probabilities are determined by the phase difference $\Delta\phi = \phi' - \phi$. From (5.45) we have

$$\Delta\phi = \arctan\left(\frac{k}{K} \tanh(Ka)\right) - \arctan\left(\frac{k}{K} \coth(Ka)\right). \quad (5.46)$$

Figure 5.12 shows the transmission probability $\sin^2(\Delta\phi)$ as a function of the energy of the incident particle for $(2mV_0a^2/\hbar^2)^{1/2} = 0.5, 1$ and 1.5 . We see that for the most permeable of these barriers the transmission probability reaches 50% when the energy is less than a third of the energy, V_0 , classically required for passage. On the other hand, the transmission probability is still only 80% when $E = V_0$ and classically the particle would be certain to pass. A barrier of the same height but three times as thick allows the particle to pass with only 2% probability when $E = V_0/3$, and even when $E = V_0$ the chance of passing this thicker barrier is only a third.

When the barrier is high, $Ka \gg 1$ so both $t = \tanh(Ka)$ and $\coth(Ka) = 1/t$ are close to unity:

$$t = \tanh(Ka) = \frac{e^{Ka} - e^{-Ka}}{e^{Ka} + e^{-Ka}} \simeq (1 - e^{-2Ka})^2. \quad (5.47)$$

Consequently, the arguments of the two arctan functions in equation (5.46) are similar and we can obtain an approximate expression for $\Delta\phi$

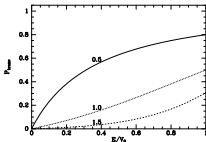


Figure 5.12 The transmission probability for a particle incident on a potential energy barrier of height V_0 and width $2a$ as a function of the particle's energy. The curves are labelled by the values of the dimensionless parameter $(2mV_0a^2/\hbar^2)^{1/2}$.

by writing

$$\begin{aligned}\arctan\left(\frac{k}{Kt}\right) &= \arctan\left(\frac{k}{Kt} + \frac{k}{Kt}(t^2 - 1)\right) \\ &\simeq \arctan\left(\frac{k}{Kt}\right) + \frac{1}{1 + (k/Kt)^2} \frac{k}{Kt}(t^2 - 1),\end{aligned}\quad (5.48)$$

where we have used the standard formula $d\arctan x/dx = 1/(1+x^2)$. Using equations (5.47) and (5.48) in equation (5.46), and we have

$$\Delta\phi \simeq -\frac{4e^{-2Ka}}{Kt/k + k/Kt} \simeq -\frac{4k}{K} e^{-2Ka} \quad (Ka \gg 1). \quad (5.49)$$

Thus the probability of passing the barrier, $\sin^2(\Delta\phi)$, decreases like e^{-4Ka} as the barrier gets higher.

5.3.2 Scattering by a classically allowed region

Now consider the case of scattering by the square potential (5.31) when $E > V_0$, so the region of non-zero potential is classically allowed. Physically that region could be a classically surmountable barrier ($V_0 > 0$) or a potential well ($V_0 < 0$). At $|x| < a$ the wavefunctions of well-defined parity are now either $\cos(Kx)$ or $A\sin(Ka)$ and from equation (5.33) the conditions for continuity of the wavefunction and its derivative at $x = a$ are

$$\left. \begin{aligned}\cos(Ka) &= B\sin(ka + \phi) \\ -K\sin(Ka) &= Bk\cos(ka + \phi)\end{aligned}\right\} \text{ or } \left\{ \begin{aligned}A\sin(Ka) &= B'\sin(ka + \phi') \\ KA\cos(Ka) &= B'k\cos(ka + \phi'),\end{aligned}\right. \quad (5.50a)$$

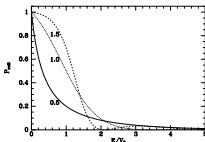


Figure 5.13 The probability of reflection by three potential barriers of height V_0 and three half-widths a as functions of E/V_0 . The curves are labelled by the dimensionless parameter $(2mV_0a^2/\hbar^2)^{1/2}$.

where

$$k = \sqrt{\frac{2mE}{\hbar^2}} \quad \text{and} \quad K = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}. \quad (5.50b)$$

By dividing the second equation of each pair into the first we obtain equations that uniquely determine the two solutions:

$$\tan(ka + \phi) = -(k/K) \cot(Ka) \quad \text{or} \quad \tan(ka + \phi') = (k/K) \tan(Ka). \quad (5.51)$$

The points in Figure 5.13 at $E > V_0$ were obtained by solving these equations⁶ for ϕ and ϕ' and then calculating the reflection probability $\cos^2(\phi' - \phi)$, while the remaining points were obtained from equations (5.46) for $E < V_0$. We see that for all three barrier widths the reflection probability obtained for $E > V_0$ joins smoothly onto that for $E < V_0$. The reflection probability tends to zero with increasing E/V_0 as we would expect, but its dependence on the thickness of the barrier is surprising: for $E/V_0 = 2$ the thickest barrier has the *lowest* reflection probability. In fact, the reflection probability vanishes for E slightly larger than $2V_0$ and then *increases* at larger energies. Similarly, the probability for transmission through the next thickest barrier vanishes near $E = 3.5V_0$. The cause of this unexpected phenomenon is quantum interference between the amplitudes to be reflected from the front and back edges of the barrier, which cancel each other when the barrier is of a particular thickness.

When the constant V_0 in the potential (5.31) is negative, there is a potential well around the origin rather than a barrier. In the classical regime the probability of reflection is zero, but as Figure 5.14 shows, it is in general non-zero and is large near $E/|V_0| \ll 1$. The oscillations in

⁶ An explicit expression (5.78) for the reflection probability in terms of ka and Ka and without reference to ϕ or ϕ' can be derived (Problem 5.10). This formula is useful when limiting cases need to be examined.

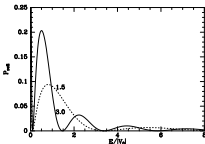


Figure 5.14 The probability for reflection by square potential wells of depth $|V_0|$. The full curve is for $(2m|V|\alpha^2/\hbar^2)^{1/2} = 3$ and the dashed curve is for a well only half as wide.

the reflection probability apparent in Figure 5.14 are caused by quantum interference between reflections from the two edges of the well.

5.3.3 Resonant scattering

In the limit that a barrier becomes very high, the probability that it reflects an incoming particle tends to unity. Consequently, a particle that encounters *two* high barriers (Figure 5.15) can bounce from one barrier to the other a great many times before eventually tunnelling through one of the barriers and escaping to infinity. This situation arises in atomic nuclei because the short-range ‘strong’ force confines charged particles such as protons and helium nuclei (α particles) within the nucleus even though it would be energetically advantageous for them to escape to infinity: the electrostatic energy released as the positively charged particle recedes from the positively charged nucleus can more than compensate for the work done on the strong force in moving beyond its short effective range (Figure 5.15). Some types of radioactivity – the sudden release of a charged particle by a nucleus – are caused by these particles tunnelling out of a well that has confined them for up to several gigayears. We now use a toy model of this physics to demonstrate that there is an important link between the cross-section for scattering by a well and the existence of long-lived bound states within the well. This connection makes it possible to probe the internal structure of atomic nuclei and ‘elementary’ particles with scattering experiments.

We model the barriers that form the potential well by δ -function potentials, located at $x = \pm a$:

$$V(x) = V_\delta \{ \delta(x+a) + \delta(x-a) \} \quad \text{with} \quad V_\delta > 0. \quad (5.52)$$



Figure 5.15 Schematic of the potential energy function $V(x)$ experienced by an α particle near an atomic nucleus. The short-range 'strong' force causes the particle's potential energy to rise extremely steeply at the edge of the nucleus. The long-range electrostatic repulsion between the nucleus and the α particle causes $V(x)$ to drop steadily as the α particle moves away from the nucleus.

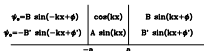


Figure 5.16 A pair of δ -function potentials form a well within which a particle can be trapped. The forms taken by the wavefunctions of the stationary states of even (top) and odd parity are shown.

By integrating the TISE

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V_\delta \delta(x)\psi = E\psi \quad (5.53)$$

for an infinitesimal distance across the location of the δ -function barriers in equation (5.52), we find that a barrier introduces a discontinuity in the gradient of the wavefunction of magnitude (cf. eq. 5.13)

$$\left[\frac{d\psi}{dx} \right] = K\psi, \quad \text{where} \quad K = \frac{2mV_\delta}{\hbar^2}. \quad (5.54)$$

Hence the energy eigenstates that will enable us to calculate scattering by a double- δ -function system take the form of sinusoids at $|x| < a$ and at $|x| > a$ that join continuously at $x = \pm a$ in such a way that their gradients there differ in accordance with equation (5.54) (Figure 5.16).

At $x = a$ the requirements on the even-parity solution $\psi_e(x)$ that it be continuous and have the prescribed change in derivative, read

$$\begin{aligned} B \sin(ka + \phi) &= \cos(ka) \\ kB \cos(ka + \phi) &= -k \sin(ka) + K \cos(ka). \end{aligned} \quad (5.55a)$$

Similarly the conditions on the odd-parity solution $\psi_o(x)$ are

$$\begin{aligned} B' \sin(ka + \phi') &= A \sin(ka) \\ kB' \cos(ka + \phi') &= kA \cos(ka) + KA \sin(ka). \end{aligned} \quad (5.55b)$$

Dividing one equation in each pair by the other to eliminate A, B and B' we obtain

$$\begin{aligned} \cot(ka + \phi) &= K/k - \tan(ka) \\ \cot(ka + \phi') &= K/k + \cot(ka). \end{aligned} \quad (5.56)$$

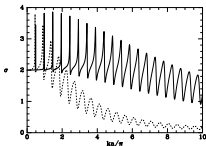


Figure 5.17 The total scattering cross-sections of double- δ -function barriers as a function of the wavenumber of the incoming particle. The barriers are located at $x = \pm a$. The full curve is for high barriers ($2mV_0a/\hbar^2 = 40$) while the dotted curve is for lower barriers ($2mV_0a/\hbar^2 = 10$).

From these expressions and equations (5.38) we can easily recover the probability $\sin^2(\phi' - \phi)$ that an incoming particle gets past both δ -function barriers. More interesting is the total scattering cross-section, which is related to the phases by equation (5.43b). Figure 5.17 shows as a function of the wavenumber of the incoming particle the cross-sections for barriers of two heights. The height of a barrier is best quantified by the dimensionless number $2mV_0a/\hbar^2 = Ka$. The full curve in Figure 5.17 is for the case that $Ka = 40$ and the dotted curve is for the case of a lower barrier such that $Ka = 10$. In each case the cross-section shows a series of peaks. In the case of the higher barrier, these peaks lie near $ka = n\pi/2$, with $n = 1, 2, \dots$. In the case of the lower barrier the peaks are less sharp and occur at slightly smaller values of ka .

If the barriers were so high as to be impenetrable, the particle would have bound states with $ka = n\pi/2$, which is the condition for the wavefunction to vanish at $x = \pm a$. Each peak in the scattering cross-section is associated with one of these bound states. Physically, the scattering cross-section is large near the energy of a bound state because at such an energy the particle can become temporarily trapped between the barriers, and after a delay escape either to the right or the left.

When the barriers have only finite height, the state $|\text{trap}\rangle$ in which the particle is initially trapped in the well is not a stationary state, and its expansion in stationary states will involve states whose energies span a non-zero range, say $(E_0 - \Gamma/2, E_0 + \Gamma/2)$. For simplicity we assume that $|\text{trap}\rangle$ has even parity, so it can be expressed as a linear combination of the even-parity stationary states $|e; E\rangle$:

$$|\text{trap}\rangle = \int_{E_0 - \Gamma/2}^{E_0 + \Gamma/2} dE a(E) |e; E\rangle, \quad (5.57)$$

where $a(E)$ is the amplitude to measure energy E . Outside the well the wavefunction of this state is

$$\psi_{\text{trap}}(x) \propto \int_{E_0-\Gamma/2}^{E_0+\Gamma/2} dE a(E) \sin(kx + \phi) \quad (x > a). \quad (5.58)$$

Below we shall find that when the well is very deep, ϕ becomes a sensitive function of E in the neighbourhood of particular 'resonant' energies. Then the sines in equation (5.58) cancel essentially perfectly on account of the rapidly changing phase $\phi(E)$. When the integral is small, there is negligible probability of finding the particle outside the well.

The evolution of ψ_{trap} with time is obtained by adding the usual factors $e^{-iEt/\hbar}$ in the integral of equation (5.58):

$$\psi_{\text{trap}}(x, t) \propto \int_{E_0-\Gamma/2}^{E_0+\Gamma/2} dE a(E) \sin(kx + \phi) e^{-iEt/\hbar} \quad (x > a). \quad (5.59)$$

After a time of order $\hbar/\Gamma$ the relative phases of the integrand at $E_0 - \Gamma/2$ and $E_0 + \Gamma/2$ will have changed by π and the originally perfect cancellation of the sines will have been sabotaged. The growth of the value of the integral, and therefore the wavefunction outside the well, signals increasing probability that the particle has escaped from the well. The more rapidly ϕ changes with E , the smaller is the value of Γ at which a negligible value of the integral in equation (5.58) can be achieved, and the smaller Γ is, the longer the particle is trapped. Thus sensitive dependence of the phases on energy is associated with long-lived trapped states, which are in turn associated with abnormally large scattering cross-sections. Notice that in Figure 5.17 the peaks are narrower at small values of k because the smaller the particle's energy is, the smaller is its probability of tunnelling through one of the barriers.

The Breit-Wigner cross-section We have seen that when particles are scattered by a model potential that contains a well, the total scattering cross-section has narrow peaks. The physical arguments given above suggest that this behaviour is generic in the sense that it is related to the time it takes a particle to tunnel out of the well after being placed there. What we have yet to do is to understand mathematically how the fairly simple formulae (5.43b) and (5.56) generate sharp peaks in the energy dependence of σ . An understanding of this phenomenon will motivate a simple analytic model of resonant scattering that is widely used in experimental physics.

Figure 5.18 shows the values of the phases ϕ and ϕ' that solve equation (5.56). For most values of k (and therefore E), the two angles are equal, so the sum $\sin^2 \phi' + \cos^2 \phi$ in equation (5.43b) is unity. The peaks in σ occur where ϕ and ϕ' briefly diverge from one another at the integral-sign features in Figure 5.18, which we shall refer to as 'glitches'.

We are interested in the case $K/k \gg 1$. Then for most values of ka the right sides of equations (5.56) are dominated by the first term, so the cotangent on the left is equal to some large positive value, and its argument lies close to zero. However each time ka/π approaches $(2r+1)\pi/2$

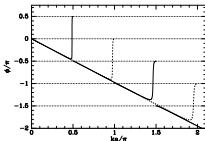


Figure 5.18 The values of the phases ϕ (full curve) and ϕ' (dashed curve) from equations (5.56) as functions of the wavenumber of the incoming particle when the latter is scattered by the double- δ -function well (eq. 5.52). These results are for the case $2mV_0a/\hbar^2 = 40$.

with r an integer, the tangent in the first equation briefly overwhelms K/k and the right side changes from a large positive number to a large negative number. Consequently the argument of the cotangent on the left quickly increases to a value close to π . As ka increases through $(2r+1)\pi/2$, the tangent instantaneously changes sign, and the argument of the cotangent instantaneously returns to a small value. Examination of the third glitch in Figure 5.18 confirms that ϕ rises rapidly but continuously by almost π and then suddenly drops by exactly π as this analysis implies. The abrupt rise in ϕ is centred on the point at which $ka + \phi = \pi/2$, at which point $\phi \simeq -r\pi$ because at a glitch $ka \simeq (2r+1)\pi/2$. Consequently, glitches in ϕ are centred on points at which $\cos^2(\phi) = 1$. Meanwhile $\phi' \simeq -(2r+1)\pi/2$, so $\sin^2(\phi') = 1$ and equation (5.43b) gives $\sigma = 4$. A very similar analysis reveals how the second of equations (5.56) generates glitches in ϕ' .

Putting this argument on a quantitative basis, we Taylor expand $\tan(ka)$ around the resonant value of k , k_R , at which $\tan(k_R a) = K/k_R$. Then

$$K/k - \tan(ka) \simeq -\sec^2(k_R a) a \delta k = -(1 + K^2/k_R^2) a \delta k, \quad (5.60)$$

where $\delta k = k - k_R$. We also observe that glitches in ϕ occur where $ka \simeq (2r+1)\pi/2$, so

$$\cot(ka + \phi) = \frac{\cos(ka) \cos(\phi) - \sin(ka) \sin \phi}{\sin(ka) \cos \phi + \cos(ka) \sin(\phi)} \simeq -\tan \phi. \quad (5.61)$$

With these approximations the first of equations (5.56) reads

$$\tan \phi \simeq (1 + K^2/k_R^2) a \delta k.$$

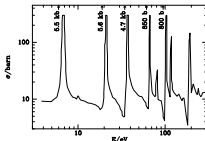


Figure 5.19 The total cross-section for the scattering of neutrons by ^{238}U nuclei. (From data published in L.M. Bollinger, *et al.*, *Phys. Rev.*, **105**, 661 (1957).)

Equation (5.43b) now gives the cross-section as

$$\begin{aligned}\sigma &= 2(\sin^2 \phi' + \cos^2 \phi) = 2\left(\sin^2 \phi' + \frac{1}{1 + \tan^2 \phi}\right) \\ &\simeq 2\left(\sin^2 \phi' + \frac{1}{1 + (1 + K^2/k_R^2)^2 a^2 (\delta k)^2}\right).\end{aligned}\quad (5.62)$$

Thus in the vicinity of the resonant energy $E_R = \hbar^2 k_R^2 / 2m$, where

$$\delta k = \frac{m}{\hbar^2 k_R}(E - E_R), \quad (5.63)$$

the cross-section has the form

$$\sigma = \text{constant} + \frac{2(\Gamma/2)^2}{(\Gamma/2)^2 + (E - E_R)^2}, \quad (5.64a)$$

where

$$\Gamma = \frac{2\hbar^2 k_R}{(1 + K^2/k_R^2)am} \simeq \frac{2\hbar^2 k_R^3}{K^2 am}. \quad (5.64b)$$

Γ has the dimensions of energy and is the characteristic width of the resonance. Experimental data for the energy dependence of cross-sections are often fitted to the functional form defined by equation (5.64a), which is known as the **Breit–Wigner cross-section**. Figure 5.19 shows a typical example.

The dependence on energy of the phase ϕ and the total scattering cross-section σ in the vicinity of a peak in σ is reminiscent of the behaviour near a resonance of a lightly damped harmonic oscillator (Box 5.1).

By the uncertainty principle, the width Γ of the Breit–Wigner cross-section (5.64a) corresponds to a timescale

$$t_R = \frac{\hbar}{\Gamma} = \frac{K^2 am}{2\hbar k_R^3}. \quad (5.65)$$

Box 5.1: Analogy with a damped oscillator

When a weakly damped harmonic oscillator is driven at some angular frequency ω , the phase of the steady-state response changes sharply in the vicinity of the oscillator's resonant frequency ω_R . Specifically, if the oscillator's equation of motion is

$$\ddot{x} + \gamma \dot{x} + \omega_R^2 x = F \cos(\omega t),$$

then the steady-state solution is $x = X \cos(\omega t - \phi)$, where

$$X = \frac{F}{\sqrt{(\omega_R^2 - \omega^2)^2 + \gamma^2 \omega^2}} \quad \text{and} \quad \phi = \arctan\left(\frac{\gamma \omega}{\omega_R^2 - \omega^2}\right).$$

As the driving frequency approaches the resonant frequency from below, the phase lag ϕ increases from near zero to $\pi/2$. As the driving frequency passes through resonance, ϕ drops discontinuously to $-\pi/2$, and then increases to near zero as $\omega^2 - \omega_R^2$ becomes large compared to $\omega\gamma$. These results suggest a picture in which a quantum well is an oscillator that is being driven by the incoming probability amplitude. The oscillator's level of damping is set by the well's characteristic energy Γ of equation (5.64b), and the form of the Breit-Wigner cross-section of equation (5.64a) mirrors the Lorentzian form of the oscillator's amplitude X .

A naive calculation confirms that t_R is the timescale on which a particle escapes from the well. When a particle encounters a δ -function barrier, it is easy to show (Problem 5.11) that its probability of tunnelling through the barrier is

$$P_{\text{tun}} = \frac{4}{4 + (K/k_R)^2} \simeq 4(k_R/K)^2 \quad \text{for } K \gg k_R. \quad (5.66)$$

Hence the probability of remaining in the well after bouncing n times off the walls is

$$P_{\text{trap}} = (1 - P_{\text{tun}})^n. \quad (5.67)$$

The particle moves from one barrier to the other in a time $t_\ell = 2am/\hbar k_R$ and in this time the logarithm of P_{trap} changes by $\ln(1 - P_{\text{tun}}) \simeq -P_{\text{tun}}$, so

$$\frac{d}{dt} \ln(P_{\text{trap}}) \simeq -\frac{P_{\text{tun}}}{t_\ell} = -\frac{1}{t_R}, \quad (5.68)$$

where t_R is given by equation (5.65). Thus this simple physical argument confirms that $\hbar/\Gamma$ is the characteristic time for the particle to remain in the well.

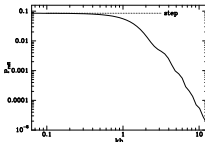


Figure 5.20 The full curve shows the probability of reflection when a particle moves from $x = -\infty$ in the potential (5.69) with energy $E = \hbar^2 k^2/2m$ and $V_0 = 0.7E$. The dotted line is the value obtained for a step change in the potential (Problem 5.4).

5.4 How applicable are our results?

It seems unlikely that any real system has a discontinuous potential $V(x)$, so our results are of practical interest only if sufficiently steep changes in V can be treated as discontinuous. We now investigate how abrupt a change in potential must be for results obtained under the assumption of a discontinuous potential to be applicable.

When the wavefunction is **evanescent** (i.e., $\propto e^{\pm Kx}$) on one side of the discontinuity, our results carry over to potentials that change continuously: where $E < V(x)$, the wavefunction is no longer a simple exponential but its phase remains constant and its amplitude decreases monotonically, while in the region $E > V(x)$ the sinusoidal dependence $\psi(x) \propto e^{ikx}$ is replaced by some other oscillatory function of similar amplitude. Qualitatively the results we have obtained for particles confined by a step potential carry over to continuously varying potentials when $E < V$ on one side of a region of varying potential.

The situation is less clear when $E > V(x)$ on both sides of the change in the potential. The relevance to such cases of our solutions for step potentials can be investigated by solving the TISE numerically for a potential that changes over a distance that can be varied (Problem 5.15). Consider for example

$$V(x) = V_0 \begin{cases} 0 & \text{for } x < -b \\ \frac{1}{2}[1 + \sin(\pi x/2b)] & \text{for } |x| < b \\ 1 & \text{for } x > b, \end{cases} \quad (5.69)$$

which changes from 0 to V_0 over a distance $2b$ centred on the origin. Figure 5.20 shows the probability of reflection when a particle with energy $\hbar^2 k^2/2m = V_0/0.7$ encounters this rise in potential energy as it approaches from $x = -\infty$. For $kb \ll 1$, the probability of reflection is close to that obtained for the corresponding step potential (Problem 5.4), but it falls

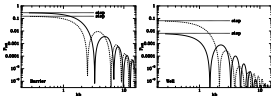


Figure 5.21 Each curve shows the reflection probability when a particle with kinetic energy E encounters a region in which the potential $V(x)$ smoothly changes to V_0 over a distance $2b$, and then smoothly returns to zero; the change in V is given by equation (5.69) with x replaced by $x + a$, and the fall is given by the same equation with x replaced by $-(x - a)$. The full curves are for $ka = 30$ and the dashed curves for $ka = 15$. The left panel is for barriers of height $V_0 = 0.7E$, while the right panel is for potential wells ($V_0 = -0.7E$).

to very much smaller values for $kb \gtrsim 2$.⁷ Thus treating a rapid change in potential energy as a discontinuity can lead to a serious overestimate of the reflected amplitude.

Figure 5.21 shows reflection probabilities for particles of energy E that encounter a finite region of elevated or depressed potential energy as a function of the sharpness of the region's sides – the changes in potential energy occur in a distance $2b$ as described by equation (5.69). The left panel is for potential barriers of height $0.7E$ and the right panel is for potential wells of depth $0.7E$. The full curves are for the case in which the region's half-width a satisfies $ka = 30$, where k is the wavenumber of the incoming and outgoing wavefunctions, while the dashed curves are for regions only half as wide. The left panel shows that when $kb = 1$, the reflection probability generated by a smooth barrier is smaller than that for a sharp step by a factor ~ 1.7 , and when $kb = 2$ it is nearly a factor ten smaller than in the case of an abrupt barrier. The right panel shows that in the case of a potential well, even smaller values of kb are required for the assumption of an abrupt change in $V(x)$ to be useful. Thus these results confirm the implication of Figure 5.20 that modelling a change in potential energy by a sharp step is seriously misleading unless the half-width of the transition region b satisfies the condition $kb < 1$.

The amplitude to be reflected by a region of varying potential energy decreases rapidly with increasing half-width b of the transition region, but the amplitudes to be reflected at the leading and trailing edges of a region of varying potential remain comparable. Consequently, destructive interference between these amplitudes is possible for all values of b . Moreover, the phases of the reflected amplitudes depend on b , so the plots of overall reflection probability versus b in Figure 5.21 show there are regular nulls in the probability for reflection like those we see in Figure 5.14 for the probability for reflection by an abrupt potential well.

⁷ The 'WKBJ' approximation derived in §12.6 provides an analytic approximation to the solution of the TISE when kb is significantly larger than 2π .

We conclude that many qualitative features of results obtained with step potentials also hold for continuous potentials, but results obtained for step potentials with no classically forbidden region are quantitatively misleading when applied to continuous potentials unless the distance over which the potential changes is small compared to the de Broglie wavelength $\lambda = h/p$ of the incident particles.

Let's consider under what circumstances this condition could be satisfied for a stream of electrons. The de Broglie wavelength of electrons with kinetic energy E is

$$\lambda = 1.16 \left(\frac{E}{1 \text{ eV}} \right)^{-1/2} \text{ nm.} \quad (5.70)$$

For the one-dimensional approximation to apply, we need the beam to be many λ wide, and for the step approximation to be valid, we require the change in potential to be complete well inside λ . In practice these conditions can be simultaneously satisfied only when the potential change is associated with a change in the medium through which the electrons are propagating. If the medium is made of atoms, the change must extend over at least the characteristic size of atoms 0.1 nm. Hence we require $E \lesssim 1 \text{ eV}$.

Realistically step potentials are relevant only for less massive particles, photons and neutrinos. The propensity for some photons to be transmitted and some reflected at an abrupt change in potential, such as that at a glass/air interface, plays an important role in optics. By contrast, electrons, neutrons and protons are unlikely to be partially transmitted and partially reflected by a region of varying potential.

These considerations explain why the phenomenon of partial reflection and partial transmission is unknown to classical mechanics, which is concerned with massive bodies that have de Broglie wavelengths many orders of magnitude smaller than an atom at any experimentally accessible energy.

5.5 Summary

In this chapter we have examined some highly idealised systems and reached some surprising conclusions.

- Any one-dimensional potential well has at least one bound state, and may have more depending on the size of the value of the dimensionless parameter W defined by equation (5.8), with V_0 and a interpreted as the well's characteristic depth and width, respectively.
- A particle trapped by a very narrow or shallow well has negligible probability of being found in the well.
- When solving the TISE in the presence of an infinite step in the potential, we should require the wavefunction to vanish at the base of the step.
- When two identical square potential wells are separated by a barrier, the eigenenergies occur in pairs, and the associated wavefunctions

have either even or odd parity with respect to an origin that is symmetrically placed between the wells. The even-parity state of a pair lies slightly lower in energy than the odd-parity state. A sum of the lowest two eigenstates is a state in which the particle is certainly in one well, while the difference gives a state in which the particle is certainly in the other well. A particle that starts in one well oscillates between the wells with a period inversely proportional to the difference between the eigenenergies. The particle is said to 'tunnel' through the barrier that divides the two wells at a rate that decreases exponentially with the product of the barrier's height and the square of its width.

- In an ammonia molecule the nitrogen atom moves in an effective potential that provides two identical wells and the above model explains how an ammonia maser works.
- A free particle has a non-zero probability to cross a potential barrier that would be impenetrable according to classical physics. On the other hand, if the potential changes significantly within one de Broglie wavelength, a particle generally has a non-zero probability of being reflected by a low barrier that classical physics predicts will be crossed.
- The probabilities for a free particle to be reflected or transmitted by a potential barrier or a well with very steep sides oscillate as functions of the particle's energy on account of quantum interference between the amplitudes to be reflected at the front and back edges of the barrier or well.
- When a free particle is scattered by a region that contains a potential well, the total scattering cross-section peaks in the vicinity of the energies of the well's approximately bound states. Longer-lived bound states are associated with sharper peaks in a plot of scattering cross-section versus energy because the width in energy of a peak, Γ , and the lifetime t_0 of the corresponding bound state are related by the uncertainty relation $t_0\Gamma \sim \hbar$.
- The Breit-Wigner formula (5.64a) gives the energy-dependence of a scattering cross-section near a resonance, and the timescale $\hbar/\Gamma$ that appears in it is the typical time for which a particle is trapped.
- The results we have obtained for discontinuous potentials $V(x)$ are an accurate guide to what will happen when the potential changes continuously when either (i) the particle is classically forbidden on one side of the change, or (ii) the change is complete within a fraction of a de Broglie wavelength. When the potential changes more gradually, the amplitude to be reflected by the region of changing potential is typically much smaller than in our idealised examples, but, as in these examples, the amplitude to be reflected oscillates as a function of energy.

Problems

5.1 A particle is confined by the potential well

$$V(x) = \begin{cases} 0 & \text{for } |x| < a \\ \infty & \text{otherwise.} \end{cases} \quad (5.71)$$

Explain (a) why we can assume that there is a complete set of stationary states with well-defined parity and (b) why to find the stationary states we solve the TISE subject to the boundary condition $\psi(\pm a) = 0$.

Determine the particle's energy spectrum and give the wavefunctions of the first two stationary states.

5.2 At $t = 0$ the particle of Problem 5.1 has the wavefunction

$$\psi(x) = \begin{cases} 1/\sqrt{2a} & \text{for } |x| < a \\ 0 & \text{otherwise.} \end{cases} \quad (5.72)$$

Find the probabilities that a measurement of its energy will yield: (a) $9\hbar^2\pi^2/(8ma^2)$; (b) $16\hbar^2\pi^2/(8ma^2)$.

5.3 Find the probability distribution of measuring momentum p for the particle described in Problem 5.2. Sketch and comment on your distribution. Hint: express $\langle p|x\rangle$ in the position representation.

5.4 Particles move in the potential

$$V(x) = \begin{cases} 0 & \text{for } x < 0 \\ V_0 & \text{for } x > 0. \end{cases} \quad (5.73)$$

Particles of mass m and energy $E > V_0$ are incident from $x = -\infty$. Show that the probability that a particle is reflected is

$$\left(\frac{k-K}{k+K} \right)^2, \quad (5.74)$$

where $k = \sqrt{2mE}/\hbar$ and $K = \sqrt{2m(E-V_0)}/\hbar$. Show directly from the TISE that the probability of transmission is

$$\frac{4kK}{(k+K)^2} \quad (5.75)$$

and check that the flux of particles moving away from the origin is equal to the incident particle flux.

5.5 Show that the energies of bound, odd-parity stationary states of the square potential well

$$V(x) = \begin{cases} 0 & \text{for } |x| < a \\ V_0 > 0 & \text{otherwise,} \end{cases} \quad (5.76)$$

are governed by

$$\cot(ka) = -\sqrt{\frac{W^2}{(ka)^2} - 1} \quad \text{where} \quad W = \sqrt{\frac{2mV_0a^2}{\hbar^2}} \quad \text{and} \quad k^2 = 2mE/\hbar^2. \quad (5.77)$$

Show that for a bound odd-parity state to exist, we require $W > \pi/2$.

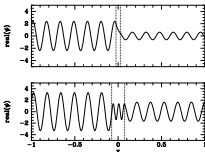


Figure 5.22 The real part of the wavefunction when a free particle of energy E is scattered by a classically forbidden square barrier (top) and a potential well (bottom). The upper panel is for a barrier of height $V_0 = E/0.7$ and half-width a such that $2mEa^2/\hbar^2 = 1$. The lower panel is for a well of depth $V_0 = E/0.2$ and half-width a such that $2mEa^2/\hbar^2 = 9$. In both panels $(2mE/\hbar^2)^{1/2} = 40$.

5.6 Show that the correctly normalised wavefunction of a particle trapped by the potential $V(x) = -V_0\delta(x)$ is $\psi(x) = \sqrt{K}e^{-K|x|}$, where $K = mV_0/\hbar^2$. Show that although this wavefunction makes it certain that a measurement of x will find the particle outside the well where its kinetic energy is negative, the expectation value of its kinetic energy $\langle E_K \rangle = \frac{1}{2}mV_0^2/\hbar^2$ is in fact positive. Reconcile this apparent paradox as follows: (i) Show that for a narrow, deep potential well of depth V_0 and half-width a , with $2V_0a = V_0$, $ka \simeq W \equiv (2mV_0a^2/\hbar^2)^{1/2}$, while $Ka \simeq W^2$. (ii) Hence show that the contribution from inside the well to $\langle E_K \rangle$ is $|\psi(0)|^2V_0$ regardless of the value of a . Explain physically what is happening as we send $a \rightarrow 0$.

5.7 Reproduce the plots shown in Figure 5.22 of the wavefunctions of particles that are scattered by a square barrier and a square potential well. Give physical interpretations of as many features of the plots as you can.

5.8 Give an example of a potential in which there is a complete set of bound stationary states of well-defined parity, and an alternative complete set of bound stationary states that are not eigenkets of the parity operator. Hint: modify the potential discussed apropos NH_3 .

5.9 A free particle of energy E approaches a square, one-dimensional potential well of depth V_0 and width $2a$. Show that the probability of being reflected by the well vanishes when $Ka = n\pi/2$, where n is an integer and $K = (2m(E+V_0)/\hbar^2)^{1/2}$. Explain this phenomenon in physical terms.

5.10 Show that the phase shifts ϕ (for the even-parity stationary state) and ϕ' (for the odd-parity state) that are associated with scattering by a

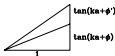


Figure 5.23 A triangle for Problem 5.10

classically allowed region of potential V_0 and width $2a$, satisfy

$$\tan(ka + \phi) = -(k/K) \cot(Ka) \quad \text{and} \quad \tan(ka + \phi') = (k/K) \tan(Ka),$$

where k and K are, respectively, the wavenumbers at infinity and in the scattering potential. Show that

$$P_{\text{refl}} = \cos^2(\phi' - \phi) = \frac{(K/k - k/K)^2 \sin^2(2Ka)}{(K/k + k/K)^2 \sin^2(2Ka) + 4 \cos^2(2Ka)}. \quad (5.78)$$

Hint: apply the cosine rule for an angle in a triangle in terms of the lengths of the triangle's sides to the top triangle in Figure 5.23.

5.11 A particle of energy E approaches from $x < 0$ a barrier in which the potential energy is $V(x) = V_0 \delta(x)$. Show that the probability of its passing the barrier is

$$P_{\text{trans}} = \frac{1}{1 + (K/2k)^2} \quad \text{where} \quad k = \sqrt{\frac{2mE}{\hbar^2}}, \quad K = \frac{2mV_0}{\hbar^2}. \quad (5.79)$$

5.12 An electron moves along an infinite chain of potential wells. For sufficiently low energies we can assume that the set $\{|n\rangle\}$ is complete, where $|n\rangle$ is the state of definitely being in the n^{th} well. By analogy with our analysis of the NH_3 molecule we assume that for all n the only non-vanishing matrix elements of the Hamiltonian are $\mathcal{E} = \langle n|H|n\rangle$ and $A = \langle n \pm 1|H|n\rangle$. Give physical interpretations of the numbers A and $\mathcal{E}$.

Explain why we can write

$$H = \sum_{n=-\infty}^{\infty} \mathcal{E}|n\rangle\langle n| + A(|n\rangle\langle n+1| + |n+1\rangle\langle n|). \quad (5.80)$$

Writing an energy eigenket $|E\rangle = \sum_n a_n |n\rangle$ show that

$$a_m(E - \mathcal{E}) - A(a_{m+1} + a_{m-1}) = 0. \quad (5.81)$$

Obtain solutions of these equations in which $a_m \propto e^{ikm}$ and thus find the corresponding energies E_k . Why is there an upper limit on the values of k that need be considered?

Initially the electron is in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|E_k\rangle + |E_{k+\Delta}\rangle), \quad (5.82)$$

where $0 < k \ll 1$ and $0 < \Delta \ll k$. Describe the electron's subsequent motion in as much detail as you can.

5.13* This problem is about the coupling of ammonia molecules to electromagnetic waves in an ammonia maser. Let $|+\rangle$ be the state in which the N atom lies above the plane of the H atoms and $|-\rangle$ be the state in which the N lies below the plane. Then when there is an oscillating electric field $\mathcal{E} \cos \omega t$ directed perpendicular to the plane of the hydrogen atoms, the Hamiltonian in the $|\pm\rangle$ basis becomes

$$H = \begin{pmatrix} \bar{E} + q\mathcal{E}s \cos \omega t & -A \\ -A & \bar{E} - q\mathcal{E}s \cos \omega t \end{pmatrix}. \quad (5.83)$$

Transform this Hamiltonian from the $|\pm\rangle$ basis to the basis provided by the states of well-defined parity $|e\rangle$ and $|o\rangle$ (where $|e\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$, etc.). Writing

$$|\psi\rangle = a_e(t)e^{-iE_e t/\hbar}|e\rangle + a_o(t)e^{-iE_o t/\hbar}|o\rangle, \quad (5.84)$$

show that the equations of motion of the expansion coefficients are

$$\begin{aligned} \frac{da_o}{dt} &= -i\Omega a_o(t) \left(e^{i(\omega - \omega_0)t} + e^{-i(\omega + \omega_0)t} \right) \\ \frac{da_e}{dt} &= -i\Omega a_e(t) \left(e^{i(\omega + \omega_0)t} + e^{-i(\omega - \omega_0)t} \right), \end{aligned} \quad (5.85)$$

where $\Omega = q\mathcal{E}s/2\hbar$ and $\omega_0 = (E_o - E_e)/\hbar$. Explain why in the case of a maser the exponentials involving $\omega + \omega_0$ can be neglected so the equations of motion become

$$\frac{da_o}{dt} = -i\Omega a_o(t)e^{i(\omega - \omega_0)t}, \quad \frac{da_e}{dt} = -i\Omega a_e(t)e^{-i(\omega - \omega_0)t}. \quad (5.86)$$

Solve the equations by multiplying the first equation by $e^{-i(\omega - \omega_0)t}$ and differentiating the result. Explain how the solution describes the decay of a population of molecules that are initially all in the higher energy level. Compare your solution to the result of setting $\omega = \omega_0$ in (5.86).

5.14 ^{238}U decays by α emission with a mean lifetime of 6.4 Gyr. Take the nucleus to have a diameter $\sim 10^{-14}$ m and suppose that the α particle has been bouncing around within it at speed $\sim c/3$. Modelling the potential barrier that confines the α particle to be a square one of height V_0 and width $2a$, give an order-of-magnitude estimate of $W = (2mV_0a^2/\hbar^2)^{1/2}$. Given that the energy released by the decay is ~ 4 MeV and the atomic number of uranium is $Z = 92$, estimate the width of the barrier through which the α particle has to tunnel. Hence give a very rough estimate of the barrier's typical height. Outline numerical work that would lead to an improved estimate of the structure of the barrier.

5.15* Particles of mass m and momentum $\hbar k$ at $x < -a$ move in the potential

$$V(x) = V_0 \begin{cases} 0 & \text{for } x < -a \\ \frac{1}{2}[1 + \sin(\pi x/2a)] & \text{for } |x| < a \\ 1 & \text{for } x > a, \end{cases} \quad (5.87)$$

where $V_0 < \hbar^2 k^2 / 2m$. Numerically reproduce the reflection probabilities plotted in Figure 5.20 as follows. Let $\psi_i = \psi(x_j)$ be the value of the wavefunction at $x_j = j\Delta$, where Δ is a small increment in the x coordinate. From the TISE show that

$$\psi_j \simeq (2 - \Delta^2 k^2) \psi_{j+1} - \psi_{j+2}, \quad (5.88)$$

where $k = \sqrt{2m(E - V)}/\hbar$. Determine ψ_j at the two grid points with the largest values of x from a suitable boundary condition, and use the recurrence relation (5.88) to determine ψ_j at all other grid points. By matching the values of ψ at the points with the smallest values of x to a sum of sinusoidal waves, determine the probabilities required for the figure. Be sure to check the accuracy of your code when $V_0 = 0$, and in the general case explicitly check that your results are consistent with equal fluxes of particles towards and away from the origin.

Equation (12.40) gives an analytical approximation for ψ in the case that there is negligible reflection. Compute this approximate form of ψ and compare it with your numerical results for larger values of a .

5.16* In this problem we obtain an analytic estimate of the energy difference between the even- and odd-parity states of a double square well. Show that for large θ , $\coth \theta - \tanh \theta \simeq 4e^{-2\theta}$. Next letting δk be the difference between the k values that solve

$$\tan[r\pi - k(b-a)] \sqrt{\frac{W^2}{(ka)^2} - 1} = \begin{cases} \coth\left(\sqrt{W^2 - (ka)^2}\right) & \text{even parity} \\ \tanh\left(\sqrt{W^2 - (ka)^2}\right) & \text{odd parity,} \end{cases} \quad (5.89a)$$

where

$$W = \sqrt{\frac{2mV_0 a^2}{\hbar^2}} \quad (5.89b)$$

for given r in the odd- and even-parity cases, deduce that

$$\left\{ \left[\left(\frac{W^2}{(ka)^2} - 1 \right)^{1/2} + \left(\frac{W^2}{(ka)^2} - 1 \right)^{-1/2} \right] (b-a) + \frac{1}{k} \left(1 - \frac{(ka)^2}{W^2} \right)^{-1} \right\} \delta k \\ \simeq -4 \exp \left[-2\sqrt{W^2 - (ka)^2} \right]. \quad (5.90)$$

Hence show that when $W \gg 1$ the fractional difference between the energies of the ground and first excited states is

$$\frac{\delta E}{E} \simeq \frac{-8a}{W(b-a)} e^{-2W\sqrt{1-E/V_0}}. \quad (5.91)$$

5.17 We consider the scattering of free particles of mass m that move in one dimension in the potential $V(x) = -W\delta(x)$, with $W > 0$. (a) For a well of finite depth V_0 and width $2a$ the condition on the phases ϕ and

ϕ' of the even- and odd-parity wavefunctions $\psi \propto \sin(kx + \phi)$, etc., for free particles are

$$\tan(ka + \phi) = -\frac{k}{K} \cot(Ka), \quad \tan(ka + \phi') = -\frac{k}{K} \tan(Ka).$$

Show that in the limit $a \rightarrow 0$, $V_0 = W/2a \rightarrow \infty$ we have $\tan \phi \rightarrow -\hbar^2 k/mW$ and $\phi' \rightarrow 0$. Hence obtain the scattering cross-section by the δ -function potential

$$\sigma = \frac{2}{1 + (\hbar k/mW)^2}. \quad (5.92)$$

(b) Re-derive the equation above for ϕ by requiring that $\psi = \sin(k|x| + \phi)$ satisfy the TISE. Convince yourself that $\psi = \sin(kx)$ is also consistent with the TISE.